Rocky Linux

리눅스 프로젝트 2026

HISTORY

조장 : 김동진
조원 : 장규혁 이영훈 정성현

Contents

Rocky Linux

01 설치

주소 설정

02 사용자 및 그룹 등록

사용자 생성

추가 그룹 생성 및 포함

03 디스크 추가 후 LVM 구성

하드디스크 추가

04 디스크 쿼터 설정

하드디스크 10G 추가

추가 사용자

05 서버 구성

SSH Server
XRDP Server
DNS Server
Web Server, FTP Server
NFS Server
Samba Server
DHCP Server
Mail Server
Maria Server

01 설치

Rocky Linux

1 Rocky Linux 9.7 설치

Vmware Workstation Pro 17

Rocky Linux 9.7 3대 설치

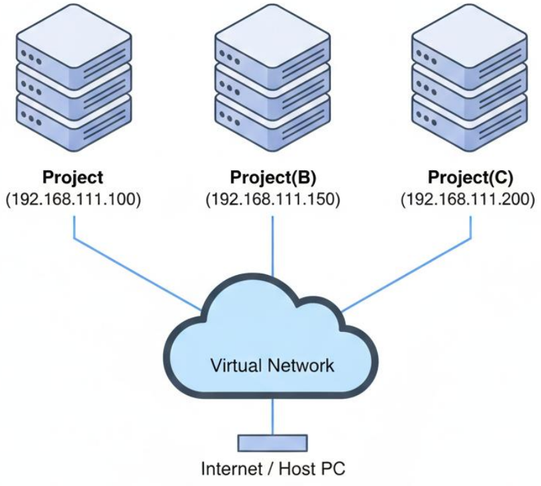
VM 이름	RAM	DISK
Project	4 GB	80 GB
Project(B)	2 GB	80 GB
Project(C)	2 GB	80 GB

2 IP address Setting

Project : 192.168.111.100/24

Project(B) : 192.168.111.150/24

Project(C) : 192.168.111.200/24



01 설치

Rocky Linux

CLI 환경에서 네트워크 변경

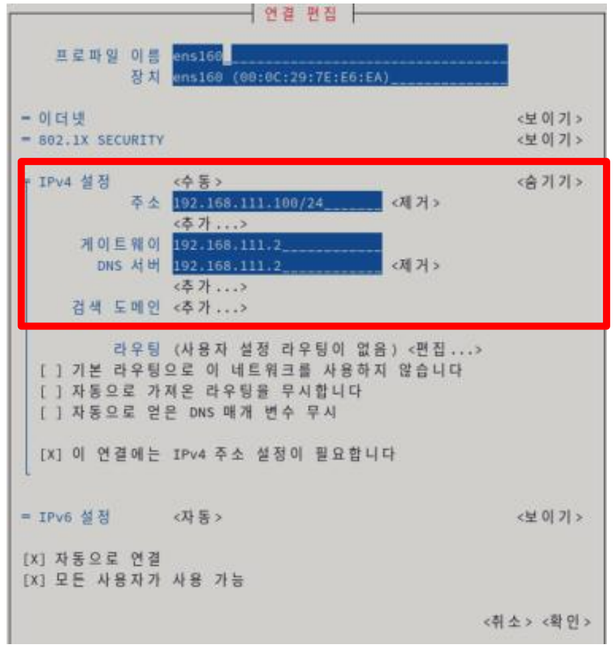
```
[root@project ~]# vi /etc/NetworkManager/system-connections/ens160.nmconnection
```

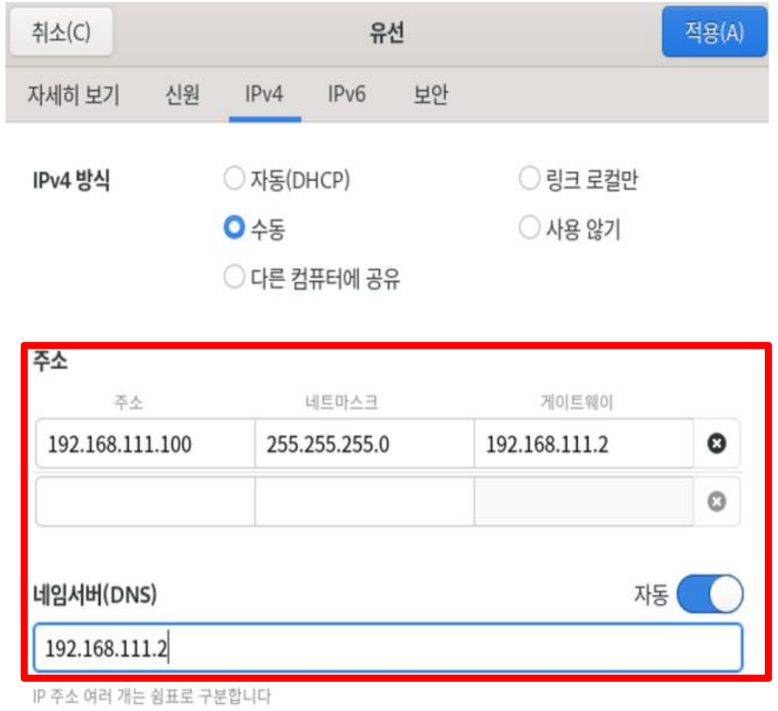
```
[connection]
id=ens160
uuid=6fbc3652-8cb3-37a0-9ced-a7623ca10332
type=ethernet
autoconnect-priority=-999
interface-name=ens160
timestamp=1767799194

[ethernet]

[ipv4]
method=auto

[ipv6]
addr-gen-mode=eui64
method=auto

[proxy]
```

```
1  [connection]
2  id=ens160
3  uuid=6fbc3652-8cb3-37a0-9ced-a7623ca10332
4  type=ethernet
5  autoconnect-priority=-999
6  interface-name=ens160
7  timestamp=1767799194
8
9  [ethernet]
10
11 [ipv4]
12 address1=192.168.111.100/24
13 dns=192.168.111.2;
14 gateway=192.168.111.2
15 method=manual
16
17 [ipv6]
18 addr-gen-mode=eui64
19 method=auto
20
21 [proxy]
```

GUI 환경에서 네트워크 변경

```
[root@project ~]# nmtui
```





01 설치 - CLI

Rocky Linux

01. 변경 전IP 주소 확인

```
[root@project ~]# ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:7e:e6:ea brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.111.136/24 brd 192.168.111.255 scope global dynamic noprefixroute ens160
       valid_lft 1659sec preferred_lft 1659sec
    inet6 fe80::20c:29ff:fe7e:e6ea/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

02. 네트워크 환경 설정

```
[root@project ~]# vi /etc/NetworkManager/system-connections/ens160.nmconnection
```

```
[connection]
id=ens160
uuid=6fbc3652-8cb3-37a0-9ced-a7623ca10332
type=ethernet
autoconnect-priority=-999
interface-name=ens160
timestamp=1767799194

[ethernet]

[ipv4]
method=auto

[ipv6]
addr-gen-mode=eui64
method=auto

[proxy]
```

```
1 [connection]
2 id=ens160
3 uuid=6fbc3652-8cb3-37a0-9ced-a7623ca10332
4 type=ethernet
5 autoconnect-priority=-999
6 interface-name=ens160
7 timestamp=1767799194
8
9 [ethernet]
10
11 [ipv4]
12 address1=192.168.111.100/24
13 dns=192.168.111.2;
14 gateway=192.168.111.2
15 method=manual
16
17 [ipv6]
18 addr-gen-mode=eui64
19 method=auto
20
21 [proxy]
```

03. 네트워크 연결 적용 및 연결

```
[root@project ~]# nmcli connection reload
[root@project ~]# nmcli connection up ens160
연결이 성공적으로 활성화되었습니다 (D-버스 활성 경로: /org/freedesktop/NetworkManager/ActiveConnection/3)
```

04. 변경 된 IP 주소 확인

```
[root@project ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:7e:e6:ea brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.111.100/24 brd 192.168.111.255 scope global noprefixroute ens160
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe7e:e6ea/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

01 설치 - GUI

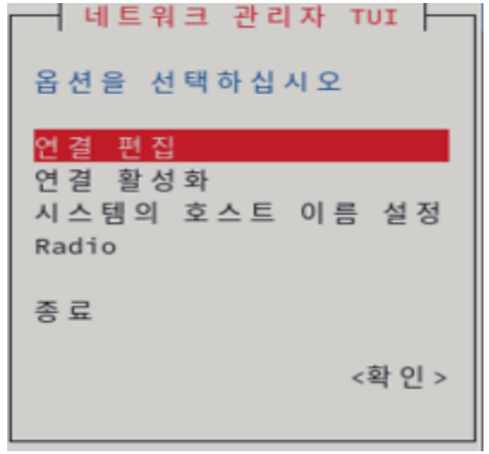
Rocky Linux

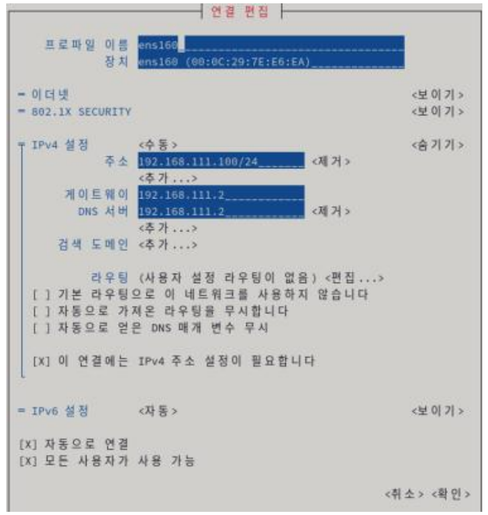
01. 변경 전IP 주소 확인

```
[root@project ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:7e:e6:ea brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.111.136/24 brd 192.168.111.255 scope global dynamic noprefixroute ens160
       valid_lft 1614sec preferred_lft 1614sec
    inet6 fe80::20c:29ff:fe7e:e6ea/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

02. 네트워크 환경 설정

```
[root@project ~]# nmtui
```





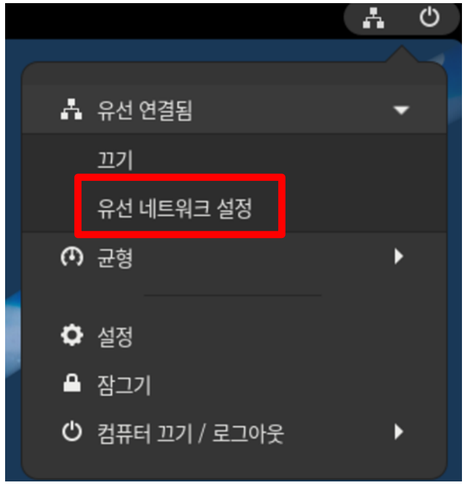
04. 변경 된 IP 주소 확인

```
[root@project ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:7e:e6:ea brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.111.100/24 brd 192.168.111.255 scope global noprefixroute ens160
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe7e:e6ea/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

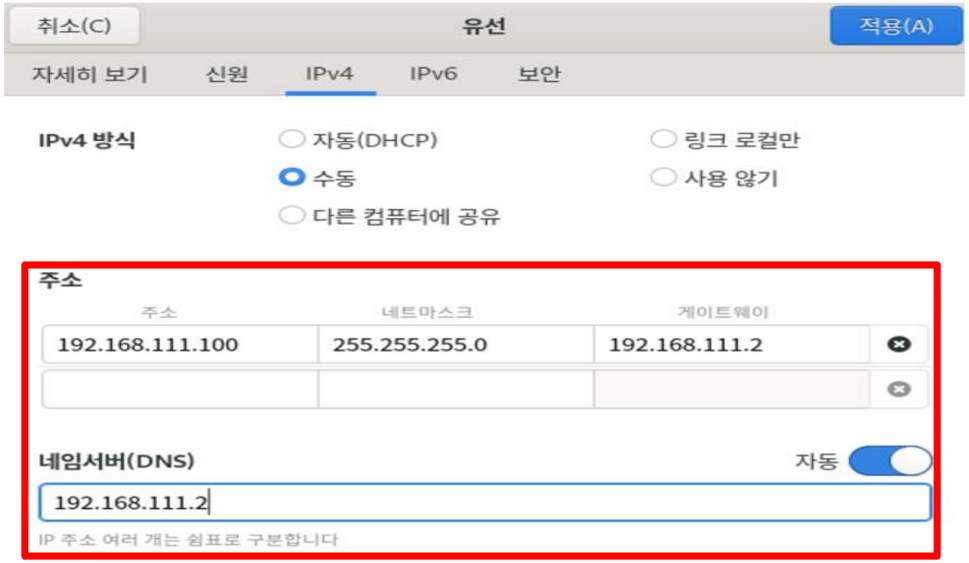
01 설치 - GUI

Rocky Linux

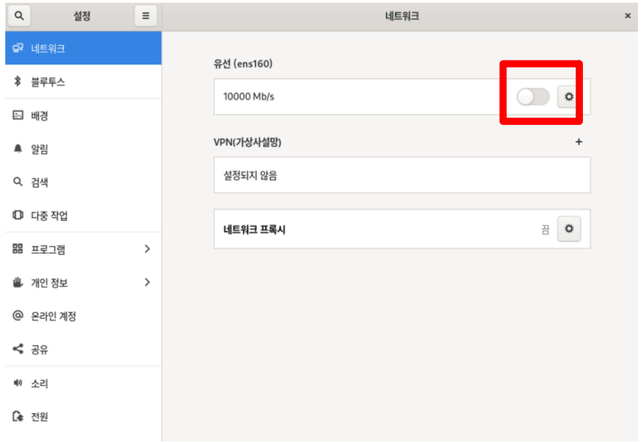
01. 유선 네트워크 설정



02. 네트워크 환경 설정



03. 네트워크 연결 해제 후 재연결



04. 변경 된 IP 주소 확인

01 설치

Rocky Linux

IP address of **project**

```
[root@project ~]# ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:7e:e6:ea brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.111.100/24 brd 192.168.111.255 scope global noprefixroute ens160
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe7e:e6ea/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

IP address of **project-b**

```
[rocky@project-b ~]$ ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:31:64:37 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.111.150/24 brd 192.168.111.255 scope global noprefixroute ens160
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe31:6437/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

IP address of **project-c**

```
[root@project-c ~]# ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:f9:b9:52 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.111.200/24 brd 192.168.111.255 scope global noprefixroute ens160
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fef9:b952/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

02 사용자 및 그룹 등록

Rocky Linux

1 사용자 생성

조원 1, 2, 3, 4 이름의 사용자를 생성한다.

손흥민, 이강인, 김민재, 황희찬 사용자를 생성한다.

2 그룹 생성

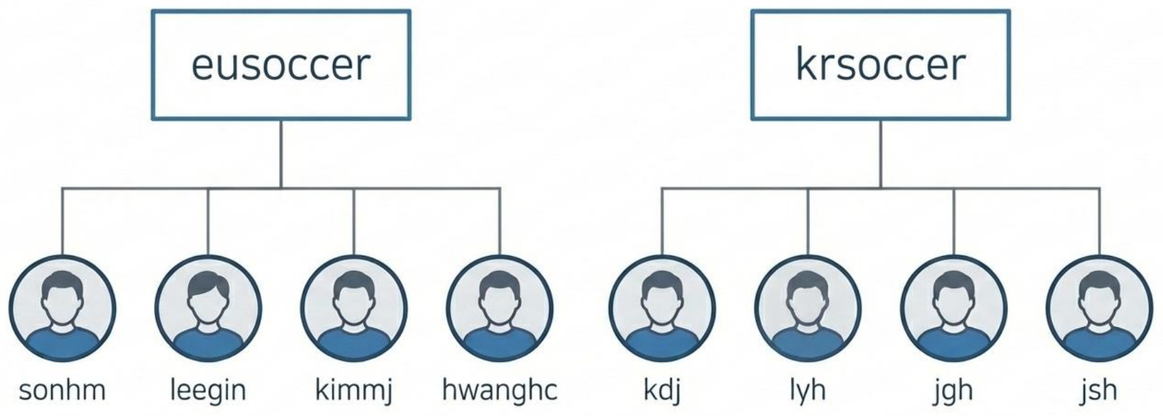
eusoccer 그룹을 생성한다.

krsoccer 그룹을 생성한다.

3 사용자를 그룹에 포함

eusoccer 그룹에 **손흥민, 이강인, 김민재, 황희찬**을 포함시킨다.

krsoccer 그룹에 **조원 1, 2, 3, 4**를 포함시킨다.



02 사용자 및 그룹 등록

Rocky Linux

01. 현재 등록된 사용자 목록

/etc/passwd 파일의 마지막 5행 출력

```
[root@project ~]# tail -5 /etc/passwd
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/usr/sbin/nologin
chrony:x:987:985:chrony system user:/var/lib/chrony:/sbin/nologin
dnsmasq:x:986:984:Dnsmasq DHCP and DNS server:/var/lib/dnsmasq:/usr/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
rocky:x:1000:1000:rocky:/home/rocky:/bin/bash
```

```
[root@project ~]# tail -5 /etc/passwd
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/usr/sbin/nologin
chrony:x:987:985:chrony system user:/var/lib/chrony:/sbin/nologin
dnsmasq:x:986:984:Dnsmasq DHCP and DNS server:/var/lib/dnsmasq:/usr/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
rocky:x:1000:1000:rocky:/home/rocky:/bin/bash
```

/etc/passwd : 사용자 계정 정보 저장 파일
계정명 : 암호 : UID : GID : 설명 : 홈디렉터리 : 로그인 쉘

02. 사용자 등록

```
[root@project ~]# adduser sonhm
[root@project ~]# adduser leegi
[root@project ~]# adduser kimmj
[root@project ~]# adduser hwanghc
[root@project ~]# adduser kdj
[root@project ~]# adduser lyh
[root@project ~]# adduser jsh
[root@project ~]# adduser jgh
```

03. 추가된 사용자 확인

기본값 - 10행 출력

```
[root@project ~]# tail /etc/passwd
tcpdump:x:72:72::/:/sbin/nologin
rocky:x:1000:1000:rocky:/home/rocky:/bin/bash
sonhm:x:1001:1001::/home/sonhm:/bin/bash
leegi:x:1002:1002::/home/leegi:/bin/bash
kimmj:x:1003:1003::/home/kimmj:/bin/bash
hwanghc:x:1004:1004::/home/hwanghc:/bin/bash
kdj:x:1005:1005::/home/kdj:/bin/bash
lyh:x:1006:1006::/home/lyh:/bin/bash
jsh:x:1007:1007::/home/jsh:/bin/bash
jgh:x:1008:1008::/home/jgh:/bin/bash
```

일반 사용자의 UID는 1000번 부터 지정된다.

02 사용자 및 그룹 등록

Rocky Linux

01. 사용자 패스워드 변경

```
[root@project ~]# passwd sonhm
sonhm 사용자의 비밀 암호 변경 중
새 암호:
잘못된 암호: 암호는 8 개의 문자 보다 짧습니다
새 암호 재입력:
passwd: 모든 인증 토큰이 성공적으로 업데이트 되었습니다.
```

패스워드는 /etc/shdow에 저장된다.
유일하게 패스워드를 지정한 sonhm의 암호만 암호화되어 표시된다.

02. 패스워드 생성 확인

```
[root@project ~]# tail /etc/shadow
tcpdump:!!:20460::::::
rocky:$6$zG3uG9Ng2zL6D56u$gv3lwzgm5MlDdAaEk/LCCd/ToKac8KiCMmv7g72Vdax.uSme5lmCDH
FeKGtlVmxc.iBOlGrLpo7Y5qygxY85t/::0:99999:7:::
sonhm:$6$rounds=100000$EonYCSMIA2mNfsNq$FtkCJQqO76UsUK445G1bCRaKHpcklvq.ldGljZyq
3lFyVIx6gtoCkHtALwmUCk.qukuYWwdLQHvet5m6XV3nc.:20460:0:99999:7:::
leegi:!!:20460:0:99999:7:::
kimmj:!!:20460:0:99999:7:::
hwanghc:!!:20460:0:99999:7:::
kdj:!!:20460:0:99999:7:::
lyh:!!:20460:0:99999:7:::
jsh:!!:20460:0:99999:7:::
jgh:!!:20460:0:99999:7:::
```

/etc/shadow 구성
계정명 : 암호 : 최종변경일 : MIN : MAX : WARNING : INACTIVE : EXPIRE : Flag

02 사용자 및 그룹 등록

Rocky Linux

01. 그룹 목록 출력

```
[root@project ~]# tail /etc/group
tcpdump:x:72:
rocky:x:1000:
sonhm:x:1001:
leegi:x:1002:
kimmj:x:1003:
hwanghc:x:1004:
kdj:x:1005:
lyh:x:1006:
jsh:x:1007:
jgh:x:1008:
```

etc/group을 통해 그룹 목록을 확인할 수 있다.

02. 그룹 생성

```
[root@project ~]# groupadd eusoccer
[root@project ~]# groupadd krsoccer
```

03. 추가된 그룹 확인

```
[root@project ~]# tail /etc/group
sonhm:x:1001:
leegi:x:1002:
kimmj:x:1003:
hwanghc:x:1004:
kdj:x:1005:
lyh:x:1006:
jsh:x:1007:
jgh:x:1008:
eusoccer:x:1009:
krsoccer:x:1010:
```

/etc/group 구성
그룹명 : 그룹암호 : GID : 그룹 멤버

02 사용자 및 그룹 등록

Rocky Linux

01. 사용자의 기본 그룹 변경

```
[root@project ~]# usermod -g eusoccer sonhm
[root@project ~]# usermod -g eusoccer leegi
[root@project ~]# usermod -g eusoccer kimmj
[root@project ~]# usermod -g eusoccer hwanghc
[root@project ~]# usermod -g krsoccer kdj
[root@project ~]# usermod -g krsoccer lyh
[root@project ~]# usermod -g krsoccer jsh
[root@project ~]# usermod -g krsoccer jgh
```

02. 변경된 기본 그룹 확인

```
[root@project ~]# tail /etc/passwd
tcpdump:x:72:72::/:/sbin/nologin
rocky x:1000:1000:rocky:/home/rocky:/bin/bash
sonhm x:1001:1009::/home/sonhm:/bin/bash
leegi x:1002:1009::/home/leegi:/bin/bash
kimmj x:1003:1009::/home/kimmj:/bin/bash
hwanghc:x:1004:1009::/home/hwanghc:/bin/bash
kdj:x:1005:1010::/home/kdj:/bin/bash
lyh:x:1006:1010::/home/lyh:/bin/bash
jsh:x:1007:1010::/home/jsh:/bin/bash
jgh:x:1008:1010::/home/jgh:/bin/bash
```

03. 그룹 멤버로 등록

```
[root@project ~]# gpasswd -a sonhm eusoccer
사용자 sonhm을(를) eusoccer 그룹에 등록 중
[root@project ~]# gpasswd -a leegi eusoccer
사용자 leegi을(를) eusoccer 그룹에 등록 중
[root@project ~]# gpasswd -a kimmj eusoccer
사용자 kimmj을(를) eusoccer 그룹에 등록 중
[root@project ~]# gpasswd -a hwanghc eusoccer
사용자 hwanghc을(를) eusoccer 그룹에 등록 중
[root@project ~]# gpasswd -a kdj krsoccer
사용자 kdj을(를) krsoccer 그룹에 등록 중
[root@project ~]# gpasswd -a lyh krsoccer
사용자 lyh을(를) krsoccer 그룹에 등록 중
[root@project ~]# gpasswd -a jsh krsoccer
사용자 jsh을(를) krsoccer 그룹에 등록 중
[root@project ~]# gpasswd -a jgh krsoccer
사용자 jgh을(를) krsoccer 그룹에 등록 중
```

04. 그룹멤버 등록 확인

```
[root@project ~]# tail /etc/group
sonhm:x:1001:
leegi:x:1002:
kimmj:x:1003:
hwanghc:x:1004:
kdj:x:1005:
lyh:x:1006:
jsh:x:1007:
jgh:x:1008:
eusoccer:x:1009:sonhm,leegi,kimmj,hwanghc
krsoccer:x:1010:kdj,lyh,jsh,jgh
```

03 디스크 추가 후 LVM 구성

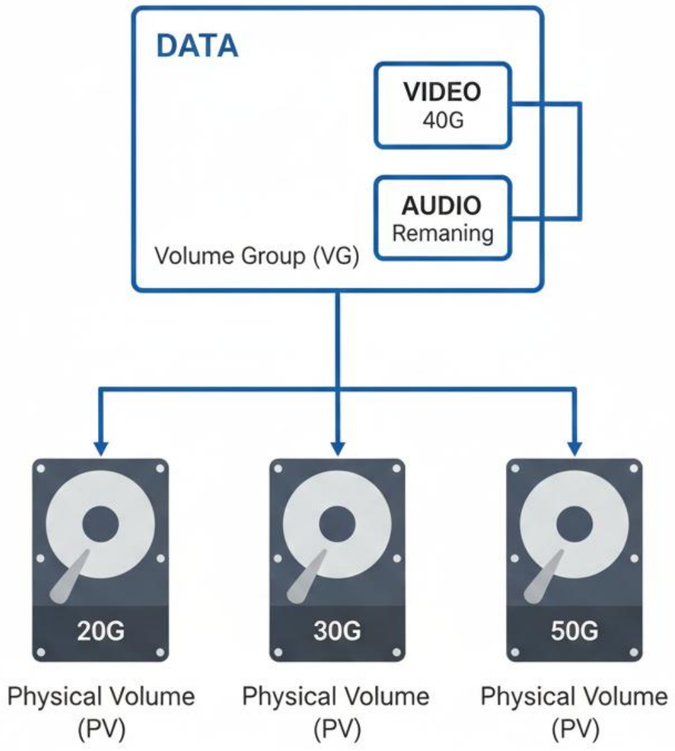
Rocky Linux

1 물리 디스크 추가

- **20GB** 물리 디스크 추가
- **30GB** 물리 디스크 추가
- **50GB** 물리 디스크 추가

2 LVM 구성

- **PV 구성**
- **VG 구성** : DATA
- **LV 구성** : VIDEO(40G), AUDIO(나머지)



03 디스크 추가 후 LVM 구성

Rocky Linux

01. lsblk를 통한 디스크 장착 확인

```
[root@project ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda           8:0    0   80G  0 disk
├sda1         8:1    0    1G  0 part /boot
└sda2         8:2    0   79G  0 part
  ├rl-root 253:0    0 50.4G  0 lvm  /
  ├rl-swap 253:1    0  3.9G  0 lvm  [SWAP]
  └rl-home 253:2    0 24.6G  0 lvm  /home
sdb           8:16   0   20G  0 disk
sdc           8:32   0   30G  0 disk
sdd           8:48   0   50G  0 disk
sr0          11:0    1 12.4G  0 rom  /run/media/root/Rocky-9-7-x86_64-dvd
```

02. fdisk를 통한 디스크 장착 확인

```
[root@project ~]# fdisk -l | grep sd
Disk /dev/sda: 80 GiB, 85899345920 bytes, 167772160 sectors
/dev/sda1   *       2048   2099199   2097152   1G 83 Linux
/dev/sda2        2099200 167772159 165672960  79G 8e Linux LVM
Disk /dev/sdb: 20 GiB, 21474836480 bytes, 41943040 sectors
Disk /dev/sdc: 30 GiB, 32212254720 bytes, 62914560 sectors
Disk /dev/sdd: 50 GiB, 53687091200 bytes, 104857600 sectors
```

03. Vmware를 통한 디스크 장착 확인

03 디스크 추가 후 LVM 구성

Rocky Linux

01. 파티션 할당

```
[root@project ~]# fdisk /dev/sdb

Device     Boot Start      End  Sectors Size Id Type
/dev/sdb1        2048 41943039 41940992  20G 8e Linux LVM
```

```
[root@project ~]# fdisk /dev/sdc

Device     Boot Start      End  Sectors Size Id Type
/dev/sdc1        2048 62914559 62912512  30G 8e Linux LVM
```

```
[root@project ~]# fdisk /dev/sdd

Device     Boot Start       End   Sectors Size Id Type
/dev/sdd1        2048 104857599 104855552  50G 8e Linux LVM
```

02. PV 구성

```
[root@project ~]# pvcreate /dev/sdb1
  Physical volume "/dev/sdb1" successfully created.
[root@project ~]# pvcreate /dev/sdc1
  Physical volume "/dev/sdc1" successfully created.
[root@project ~]# pvcreate /dev/sdd1
  Physical volume "/dev/sdd1" successfully created.
```

```
[root@project ~]# pvscan
  PV /dev/sda2   VG rl     lvm2 [<79.00 GiB / 0    free]
  PV /dev/sdb1   VG DATA   lvm2 [<20.00 GiB / <20.00 GiB free]
  PV /dev/sdc1   VG DATA   lvm2 [<30.00 GiB / <30.00 GiB free]
  PV /dev/sdd1   VG DATA   lvm2 [<50.00 GiB / <50.00 GiB free]
  Total: 4 [178.98 GiB] / in use: 4 [178.98 GiB] / in no VG: 0 [0   ]
```

03. VG 구성

```
[root@project ~]# vgcreate DATA /dev/sdb1 /dev/sdc1 /dev/sdd1
  Volume group "DATA" successfully created
```

```
[root@project ~]# vgdisplay
  --- Volume group ---
  VG Name               DATA
  System ID
  Format                lvm2
  Metadata Areas        3
  Metadata Sequence No  1
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                0
  Open LV               0
  Max PV                0
  Cur PV                3
  Act PV                3
  VG Size               <99.99 GiB
  PE Size               4.00 MiB
  Total PE              25597
  Alloc PE / Size       0 / 0
  Free  PE / Size       25597 / <99.99 GiB
  VG UUID               WRB7Ex-dtBE-8AtT-mll7-5zmp-YOcC-E1Uqq2
```

04. LV 구성

```
[root@project ~]# lvcreate --size 40G --name VIDEO DATA
  Logical volume "VIDEO" created.
[root@project ~]# lvcreate --extents 100%FREE --name AUDIO DATA
  Logical volume "AUDIO" created.
```

```
[root@project ~]# lvscan
  ACTIVE            '/dev/rl/swap' [3.91 GiB] inherit
  ACTIVE            '/dev/rl/home' [24.63 GiB] inherit
  ACTIVE            '/dev/rl/root' [<50.45 GiB] inherit
  ACTIVE            '/dev/DATA/VIDEO' [40.00 GiB] inherit
  ACTIVE            '/dev/DATA/AUDIO' [<59.99 GiB] inherit
```

```
[root@project ~]# ls -l /dev/DATA
합계 0
lrwxrwxrwx. 1 root root 7  1월  8 15:14 AUDIO -> ../dm-4
lrwxrwxrwx. 1 root root 7  1월  8 15:13 VIDEO -> ../dm-3
```

03 디스크 추가 후 LVM 구성

Rocky Linux

01. 파일 시스템 생성

```
[root@project ~]# mkfs.ext4 /dev/DATA/VIDEO
mke2fs 1.46.5 (30 [illegible])
Creating filesystem with 10485760 4k blocks and 2621440 inodes
Filesystem UUID: b5cc662a-08c5-4122-8e46-6a3e493ff176
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
        4096000, 7962624

Allocating group tables: done
Writing inode tables: done
Creating journal (65536 blocks): done
Writing superblocks and filesystem accounting information: done
```

```
[root@project ~]# mkfs.ext4 /dev/DATA/AUDIO
mke2fs 1.46.5 (30 [illegible])
Creating filesystem with 15725568 4k blocks and 3932160 inodes
Filesystem UUID: 7e94002f-668b-445c-ae21-6c9064b3f334
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
        4096000, 7962624, 11239424

Allocating group tables: done
Writing inode tables: done
Creating journal (65536 blocks): done
Writing superblocks and filesystem accounting information: done
```

02. 마운트

```
[root@project ~]# mkdir /lvm1 /lvm2
```

```
[root@project ~]# mount /dev/DATA/VIDEO /lvm1
[root@project ~]# mount /dev/DATA/AUDIO /lvm2
```

```
[root@project ~]# df
Filesystem             1K-blocks     Used Available Use% Mounted on
devtmpfs                    4096        0      4096   0% /dev
tmpfs                    1854880        0   1854880   0% /dev/shm
tmpfs                     741956     9944    732012   2% /run
/dev/mapper/rl-root     52834304  6439648  46394656  13% /
/dev/sda1                 983040   382744    600296  39% /boot
/dev/mapper/rl-home     25763840   212896  25550944   1% /home
tmpfs                     370976      104    370872   1% /run/user/0
/dev/sr0                13047408 13047408         0 100% /run/media/root/Rocky-9
-7-x86_64-dvd
/dev/mapper/DATA-VIDEO  40973536       24  38859976   1% /lvm1
/dev/mapper/DATA-AUDIO  61599532       24  58438012   1% /lvm2
```

03. /etc/fstab 설정

```
[root@project ~]# vi /etc/fstab
```

```
#
# /etc/fstab
# Created by anaconda on Wed Jan  7 06:22:54 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
/dev/mapper/rl-root     /                       xfs     defaults        0 0
UUID=89edbb41-ed45-48d6-9040-383ce190daaf /boot                   xfs     defaul
ts        0 0
/dev/mapper/rl-home     /home                   xfs     defaults        0 0
/dev/mapper/rl-swap     none                    swap    defaults        0 0
/dev/DATA/VIDEO /lvm1   ext4    defaults        0       0
/dev/DATA/AUDIO /lvm2   ext4    defaults        0       0
```

04 디스크 쿼터 설정

Rocky Linux

1 물리 디스크 추가

- **10GB** 하드 디스크 추가

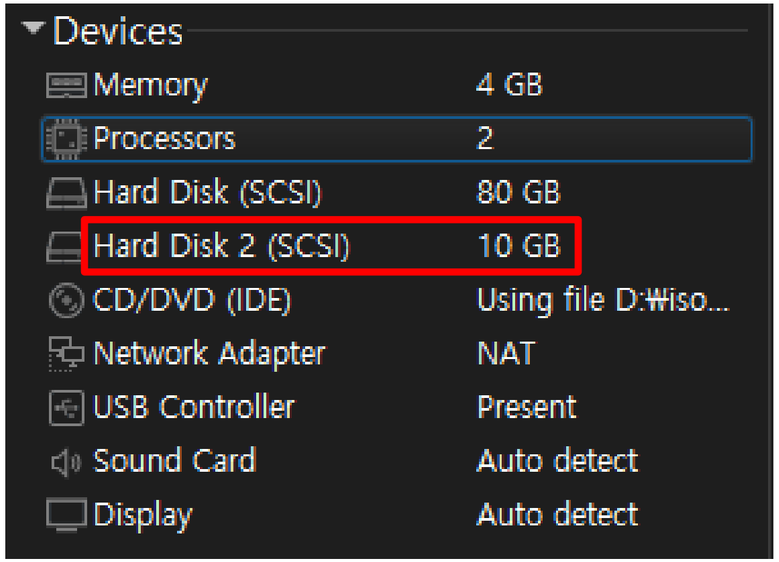
2 추가 사용자

- **aespa** - soft 700M | hard 1G
- **IVE** - soft 700M | hard 1G
- **NewJeans** - soft 700M | hard 1G

04 디스크 쿼터 설정

Rocky Linux

01. Vmware를 통한 디스크 장착 확인



02. lsblk를 통한 디스크 장착 확인

```
[root@project ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda           8:0    0   80G  0 disk
├sda1         8:1    0    1G  0 part /boot
└sda2         8:2    0   79G  0 part
  ├rl-root 253:0    0 50.4G  0 lvm  /
  ├rl-swap 253:1    0  3.9G  0 lvm  [SWAP]
  └rl-home 253:2    0 24.6G  0 lvm  /home
sdb           8:16   0   10G  0 disk
sr0          11:0    1 12.4G  0 rom  /run/media/root/Rocky-9-7-x86_64-dvd
```

03. 파티션 생성 및 마운트

```
[root@project ~]# fdisk /dev/sdb

Device     Boot Start      End  Sectors Size Id Type
/dev/sdb1          2048 20971519 20969472  10G 83 Linux

[root@project ~]# mkfs.ext4 /dev/sdb1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 2621184 4k blocks and 655360 inodes
Filesystem UUID: f7f37e0b-468c-4d74-ad64-d86aaf67d47f
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

[root@project ~]# mkdir /userHome
[root@project ~]# mount /dev/sdb1 /userHome
```

04. /etc/fstab 수정

```
[root@project ~]# vi /etc/fstab

#
# /etc/fstab
# Created by anaconda on Wed Jan  7 06:22:54 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
/dev/mapper/rl-root     /                       xfs     defaults        0 0
UUID=89edbb41-ed45-48d6-9040-383ce190daaf /boot                   xfs     defaul
ts        0 0
/dev/mapper/rl-home     /home                   xfs     defaults        0 0
/dev/mapper/rl-swap     none                    swap    defaults        0 0
/dev/sdb1       /userHome       ext4    defaults        0       0
```

04 디스크 쿼터 설정

Rocky Linux

01. 사용자 및 암호 생성

```
[root@project ~]# useradd -d /userHome/aespa aespa
[root@project ~]# useradd -d /userHome/IVE IVE
[root@project ~]# useradd -d /userHome/NewJeans NewJeans
```

```
[root@project ~]# passwd aespa
aespa 사용자의 비밀 번호 변경 중
새 암호:
잘못된 암호: 암호는 8 개의 문자 보다 짧습니다
새 암호 재입력:
passwd: 모든 인증 토큰이 성공적으로 업데이트 되었습니다.
```

```
[root@project ~]# passwd IVE
IVE 사용자의 비밀 번호 변경 중
새 암호:
잘못된 암호: 암호는 8 개의 문자 보다 짧습니다
새 암호 재입력:
passwd: 모든 인증 토큰이 성공적으로 업데이트 되었습니다.
```

```
[root@project ~]# passwd NewJeans
NewJeans 사용자의 비밀 번호 변경 중
새 암호:
잘못된 암호: 암호는 8 개의 문자 보다 짧습니다
새 암호 재입력:
passwd: 모든 인증 토큰이 성공적으로 업데이트 되었습니다.
```

02. /etc/fstab 수정

```
[root@project ~]# vi /etc/fstab
```

```
#
# /etc/fstab
# Created by anaconda on Wed Jan  7 06:22:54 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
/dev/mapper/rl-root     /                       xfs     defaults        0 0
UUID=89edbb41-ed45-48d6-9040-383ce190daaf /boot                   xfs     defaults        0 0
/dev/mapper/rl-home     /home                   xfs     defaults        0 0
/dev/mapper/rl-swap     none                    swap    defaults        0 0
/dev/sdb1       /userHome       ext4    defaults,usrjquota=aquota.user,jqfmt=vfsv0      0       0
```

03. remount를 통한 재부팅 효과

```
[root@project ~]# mount --options remount /userHome
mount: (hint) your fstab has been modified, but systemd still uses
       the old version; use 'systemctl daemon-reload' to reload.
```

```
[root@project ~]# mount
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime,seclabel)
```

04 디스크 쿼터 설정

Rocky Linux

01. 쿼터 DB 생성

```
[root@project ~]# cd /userHome
[root@project userHome]# quotaoff -avug
quotaoff: Your kernel probably supports ext4 quota feature but you are using external quota files.
 Please switch your filesystem to use ext4 quota feature as external quota files on ext4 are depre
cated. You can enable the feature by unmounting the file system and running 'tune2fs -O quota <dev
ice>'.
/dev/sdb1 [/userHome]: user quotas turned off
[root@project userHome]# quotacheck -augmn
quotacheck: Your kernel probably supports ext4 quota feature but you are using external quota file
s. Please switch your filesystem to use ext4 quota feature as external quota files on ext4 are dep
recated. You can enable the feature by unmounting the file system and running 'tune2fs -O quota <d
evice>'.
[root@project userHome]# rm -rf aquota.*
[root@project userHome]# quotacheck -augmn
quotacheck: Your kernel probably supports ext4 quota feature but you are using external quota file
s. Please switch your filesystem to use ext4 quota feature as external quota files on ext4 are dep
recated. You can enable the feature by unmounting the file system and running 'tune2fs -O quota <d
evice>'.
[root@project userHome]# touch aquota.user aquota.group
[root@project userHome]# chmod 600 aquota.*
[root@project userHome]# quotacheck -augmn
quotacheck: Your kernel probably supports ext4 quota feature but you are using external quota file
s. Please switch your filesystem to use ext4 quota feature as external quota files on ext4 are dep
recated. You can enable the feature by unmounting the file system and running 'tune2fs -O quota <d
evice>'.
[root@project userHome]# quotaon -avug
quotaon: Your kernel probably supports ext4 quota feature but you are using external quota files.
Please switch your filesystem to use ext4 quota feature as external quota files on ext4 are deprec
ated. You can enable the feature by unmounting the file system and running 'tune2fs -O quota <devi
ce>'.
/dev/sdb1 [/userHome]: user quotas turned on
```

02. 사용자별 사용 공간 할당

root@project:/userHome — edquota -u aespa

```
Disk quotas for user aespa (uid 1009):
  Filesystem                  blocks       soft       hard     inodes     soft     hard
  /dev/sdb1                       28        700       1000          7        0        0
```

root@project:/userHome — edquota -u IVE

```
Disk quotas for user IVE (uid 1010):
  Filesystem                  blocks       soft       hard     inodes     soft     hard
  /dev/sdb1                       28        700       1000          7        0        0
```

root@project:/userHome — edquota -u NewJeans

```
Disk quotas for user NewJeans (uid 1011):
  Filesystem                  blocks       soft       hard     inodes     soft     hard
  /dev/sdb1                       28        700       1000          7        0        0
```

04 디스크 쿼터 설정

Rocky Linux

01. 사용자별 사용 공간 확인

```
[root@project userHome]# su - aespa
[aespa@project ~]$ pwd
/userHome/aespa
[aespa@project ~]$ cp /boot/vmlinuz-5* test1
sdb1: warning, user block quota exceeded.
sdb1: write failed, user block limit reached.
cp: 'test1'에 쓰는 도중 오류 발생: 디스크 할당량이 초과됨
[aespa@project ~]$ cp test1 test2
cp: error copying 'test1' to 'test2': 디스크 할당량이 초과됨
[aespa@project ~]$ cp test1 test3
cp: error copying 'test1' to 'test3': 디스크 할당량이 초과됨
[aespa@project ~]$ ls -l
합계 968
-rwxr-xr-x. 1 aespa aespa 991232  1월  8 15:49 test1
-rwxr-xr-x. 1 aespa aespa      0  1월  8 15:50 test2
-rwxr-xr-x. 1 aespa aespa      0  1월  8 15:50 test3
[aespa@project ~]$ cp test1 test4
cp: error copying 'test1' to 'test4': 디스크 할당량이 초과됨
[aespa@project ~]$ ls -l
합계 968
-rwxr-xr-x. 1 aespa aespa 991232  1월  8 15:49 test1
-rwxr-xr-x. 1 aespa aespa      0  1월  8 15:50 test2
-rwxr-xr-x. 1 aespa aespa      0  1월  8 15:50 test3
-rwxr-xr-x. 1 aespa aespa      0  1월  8 15:50 test4
```

02. quota 명령으로 디스크 공간 확인

```
[aespa@project ~]$ quota
Disk quotas for user aespa (uid 1009):
     Filesystem  blocks   quota   limit   grace   files   quota   limit   grace
      /dev/sdb1   1000*    700    1000   6days      12       0       0
```

```
[IVE@project ~]$ quota
Disk quotas for user IVE (uid 1010):
     Filesystem  blocks   quota   limit   grace   files   quota   limit   grace
      /dev/sdb1   1000*    700    1000   6days      12       0       0
```

```
[NewJeans@project ~]$ quota
Disk quotas for user NewJeans (uid 1011):
     Filesystem  blocks   quota   limit   grace   files   quota   limit   grace
      /dev/sdb1   1000*    700    1000   6days      12       0       0
```

04 디스크 쿼터 설정

Rocky Linux

01. repquota /userHome 명령으로 현재 사용량 확인

```
[aespa@project ~]$ exit
로그아웃
[root@project userHome]# repquota /userHome
*** Report for user quotas on device /dev/sdb1
Block grace time: 7days; Inode grace time: 7days
                        Block limits                File limits
User            used    soft    hard  grace    used  soft  hard  grace
----------------------------------------------------------------------
root      --      20       0       0              3     0     0
aespa     +-    1000     700    1000  6days      13     0     0
IVE       --      28     700    1000               7     0     0
NewJeans  --      28     700    1000               7     0     0
```

```
[IVE@project ~]$ exit
로그아웃
[root@project userHome]# repquota /userHome
*** Report for user quotas on device /dev/sdb1
Block grace time: 7days; Inode grace time: 7days
                        Block limits                File limits
User            used    soft    hard  grace    used  soft  hard  grace
----------------------------------------------------------------------
root      --      20       0       0              3     0     0
aespa     +-    1000     700    1000  6days      13     0     0
IVE       +-    1000     700    1000  6days      13     0     0
NewJeans  --      28     700    1000               7     0     0
```

```
[NewJeans@project ~]$ exit
로그아웃
[root@project userHome]# repquota /userHome
*** Report for user quotas on device /dev/sdb1
Block grace time: 7days; Inode grace time: 7days
                        Block limits                File limits
User            used    soft    hard  grace    used  soft  hard  grace
----------------------------------------------------------------------
root      --      20       0       0              3     0     0
aespa     +-    1000     700    1000  6days      13     0     0
IVE       +-    1000     700    1000  6days      13     0     0
NewJeans  +-    1000     700    1000  6days      13     0     0
```

02. 사용자에게 적용된 사항을 다른 사용자에게 적용

```
[root@project userHome]# edquota -p aespa IVE
[root@project userHome]# repquota /userHome
*** Report for user quotas on device /dev/sdb1
Block grace time: 7days; Inode grace time: 7days
                        Block limits                File limits
User            used    soft    hard  grace    used  soft  hard  grace
----------------------------------------------------------------------
root      --      20       0       0              3     0     0
aespa     +-    1000     700    1000  6days      13     0     0
IVE       --      28     700    1000               7     0     0
NewJeans  --      28     700    1000               7     0     0
```

```
[root@project userHome]# edquota -p IVE NewJeans
[root@project userHome]# repquota /userHome
*** Report for user quotas on device /dev/sdb1
Block grace time: 7days; Inode grace time: 7days
                        Block limits                File limits
User            used    soft    hard  grace    used  soft  hard  grace
----------------------------------------------------------------------
root      --      20       0       0              3     0     0
aespa     +-    1000     700    1000  6days      13     0     0
IVE       +-    1000     700    1000  6days      13     0     0
NewJeans  --      28     700    1000               7     0     0
```

05 서버 구성

Rocky Linux

1 서버 구성

- **SSH** Server
- **XRDP** Server
- **DNS** Server
- **Web** Server, **FTP** Server
- **NFS** Server
- **Samba** Server
- **DHCP** Server
- **Mail** Server
- **Maria** Server

05 서버 구성 - SSH

Rocky Linux

01. SSH 설치 여부 확인 및 설치

```
[root@project ~]# rpm -qa openssh-server
openssh-server-8.7p1-46.el9.rocky.0.1.x86_64
[root@project ~]# dnf install openssh-server
```

미설치시 명령어 실행

02. 환경설정

```
[root@project ~]# vi /etc/ssh/sshd_config
```

필요시 설정변경

03. 서비스 시작

```
[root@project ~]# systemctl restart sshd
[root@project ~]# systemctl enable sshd
```

04. SSH 서비스 방화벽 허가

```
[root@project ~]# firewall-cmd --permanent --add-service=ssh
success
[root@project ~]# firewall-cmd --reload
success
[root@project ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns imap mysql nfs pop3 samba samba-client smtp ssh
```

05. XShell 연결

일반

이름(N):	project
프로토콜(P):	SSH
호스트(H):	192.168.111.100
포트 번호(O):	22

```
Xshell 8 (Build 0084)
Copyright (c) 2024 NetSarang Computer, Inc. All rights reserved.

Type `help' to learn how to use Xshell prompt.
[C:\~]$

Connecting to 192.168.111.100:22...
Connection established.
To escape to local shell, press 'Ctrl+Alt+]'.

Activate the web console with: systemctl enable --now cockpit.socket

Last login: Tue Jan 13 15:01:35 2026
[root@project ~]#
```

05 서버 구성 - XRDP

Rocky Linux

01. XRDP 설치 여부 확인 및 설치

```
[root@project ~]# rpm -qa xrdp
[root@project ~]# dnf -y install epel-release
```

02. 서비스 시작 및 활성화

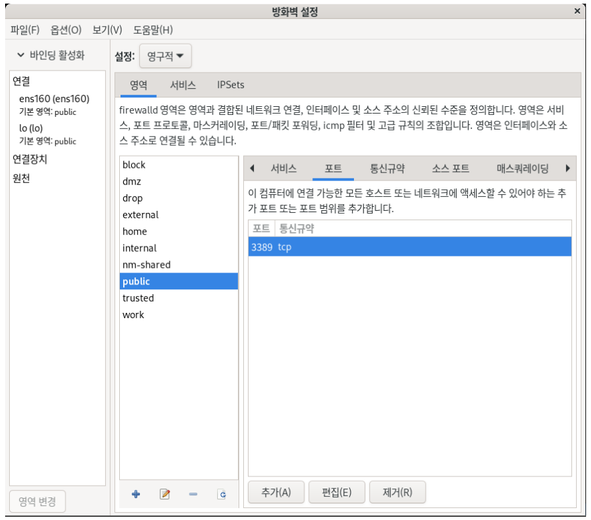
```
[root@project ~]# systemctl start xrdp
[root@project ~]# systemctl enable xrdp
[root@project ~] systemctl status xrdp
● xrdp.service - xrdp daemon
   Loaded: loaded (/usr/lib/systemd/system/xrdp.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-01-08 16:08:25 KST; 22s ago
```

03. CLI를 통한 방화벽 허가

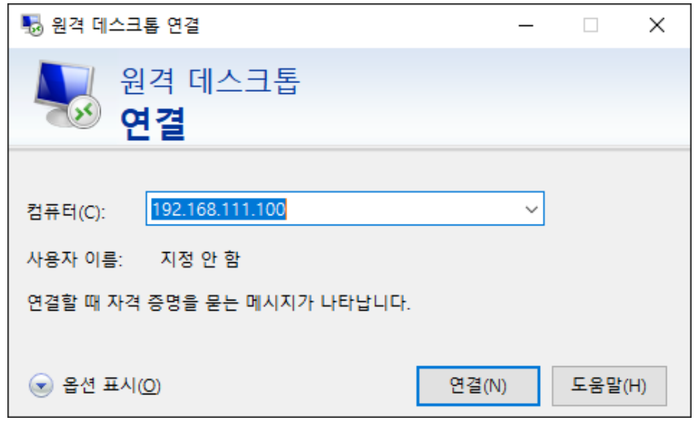
```
[root@project ~]# firewall-cmd --permanent --add-port=3389/tcp
success
[root@project ~]# firewall-cmd --reload
success
[root@project ~]# firewall-cmd --list-ports
3389/tcp
```

04. GUI를 통한 방화벽 허가

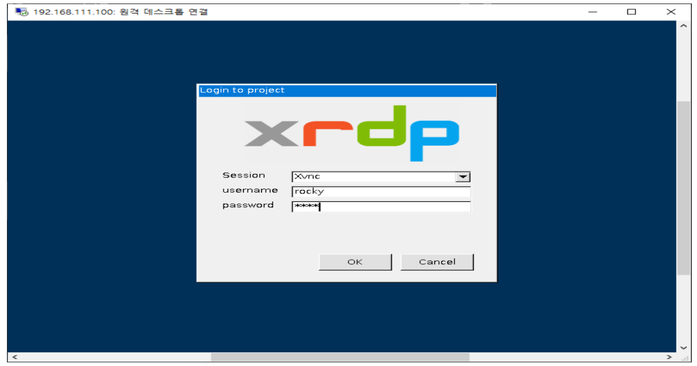
```
[root@project ~]# firewall-config
```

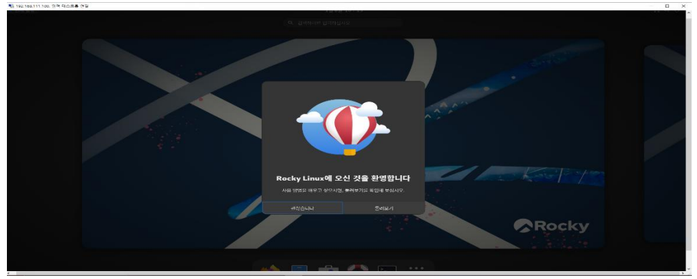


05. Project의 IP주소 입력 후 연결



06. 계정과 비밀번호 입력 후 접속





05 서버 구성 - DNS

Rocky Linux

01. DNS 설치 여부 확인 및 설치

```
[root@project ~]# rpm -qa bind bind-chroot
[root@project ~]# dnf -y install bind bind-chroot
```

02. 환경설정

```
[root@project ~]# vi /etc/named.conf
```

```
10 options {
11         listen-on port 53 { any; };
12         listen-on-v6 port 53 { none; };
13         directory       "/var/named";
14         dump-file       "/var/named/data/cache_dump.db";
15         statistics-file "/var/named/data/named_stats.txt";
16         memstatistics-file "/var/named/data/named_mem_stats.txt";
17         secroots-file   "/var/named/data/named.secroots";
18         recursing-file  "/var/named/data/named.recursing";
19         allow-query     { any; };
20
21         /*
22          - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
23          - If you are building a RECURSIVE (caching) DNS server, you need to enable
24            recursion.
25          - If your recursive DNS server has a public IP address, you MUST enable access
26            control to limit queries to your legitimate users. Failing to do so will
27            cause your server to become part of large scale DNS amplification
28            attacks. Implementing BCP38 within your network would greatly
29            reduce such attack surface
30         */
31         recursion yes;
32
33         dnssec-validation no;
```

03. 서비스 시작

```
[root@project ~]# systemctl start named
[root@project ~]# systemctl enable named
Created symlink /etc/systemd/system/multi-user.target.wants/named.service → /usr/lib/systemd/
system/named.service.
[root@project ~]# systemctl status named
● named.service - Berkeley Internet Name Domain (DNS)
     Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: disabled)
     Active: active (running) since Fri 2026-01-09 15:05:57 KST; 17s ago
   Main PID: 6491 (named)
      Tasks: 8 (limit: 22762)
     Memory: 26.8M (peak: 27.5M)
        CPU: 78ms
     CGroup: /system.slice/named.service
             └─6491 /usr/sbin/named -u named -c /etc/named.conf
```

04. 방화벽 허가

```
[root@project ~]# firewall-cmd --permanent --add-service=dns
success
[root@project ~]# firewall-cmd --reload
success
[root@project ~]# firewall-cmd --list-services
cockpit dhcpv6-client dns ssh
```

05. DNS 서버 작동 확인

```
[root@project ~]# nslookup
> server 192.168.111.100
Default server: 192.168.111.100
Address: 192.168.111.100#53
> www.nate.com
Server:         192.168.111.100
Address:        192.168.111.100#53

Non-authoritative answer:
Name:   www.nate.com
Address: 120.50.131.112
```

05 서버 구성 - DNS

Rocky Linux

06. 도메인 설정 후 오류 검사

```
[root@project-b ~]# vi /etc/resolv.conf
# Generated by NetworkManager
nameserver 192.168.111.100
```

```
60 zone "history.com" IN {
61         type master;
62         file "history.com.db";
63         allow-update { none; };
64 };
```

```
[root@project ~]# named-checkconf
[root@project ~]#
```

07. 도메인 존 파일 설정

```
[root@project named]# vi history.com.db
```

```
1 $TTL    3H
2 @       SOA     @       root.   ( 2     1D      1H      1W      1H )
3         IN      NS      @
4         IN      A       192.168.111.100
5
6 proeject        IN      A       192.168.111.100
7 project-b       IN      A       192.168.111.150
8 project-c       IN      A       192.168.111.200
9
10
11 www    IN      CNAME   project-b
12 ftp    IN      CNAME   project-c
```

08. 존 파일 오류 검사

```
[root@project named]# named-checkzone history.com history.com.db
zone history.com/IN: loaded serial 2
OK
```

문법에 문제가 있는지 체크한다.

09. 서버 접속 확인

```
[root@project named]# nslookup
> www.history.com
Server:         192.168.111.100
Address:        192.168.111.100#53

www.history.com canonical name = project-b.history.com.
Name:   project-b.history.com
Address: 192.168.111.150
> ftp.history.com
Server:         192.168.111.100
Address:        192.168.111.100#53

ftp.history.com canonical name = project-c.history.com.
Name:   project-c.history.com
Address: 192.168.111.200
```

www은 project-b에 대한 CNAME 레코드로 설정
www.history.com 조회 시 project-b가 가리키는 IP 주소 반환

ftp는 project-에 대한 CNAME 레코드로 설정
ftp.history.com 조회 시 project-c가 가리키는 IP주소 반환

05 서버 구성 - Web

Rocky Linux

01. Web Server 설치 여부 확인 및 설치

```
[root@project-b ~]# rpm -qa httpd
[root@project-b ~]# dnf install -y httpd
```

02. 서비스 시작

```
[root@project-b ~]# systemctl start httpd
[root@project-b ~]# systemctl enable httpd
```

```
[root@project-b ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Fri 2026-01-09 16:13:55 KST; 14s ago
```

03. 방화벽 허가

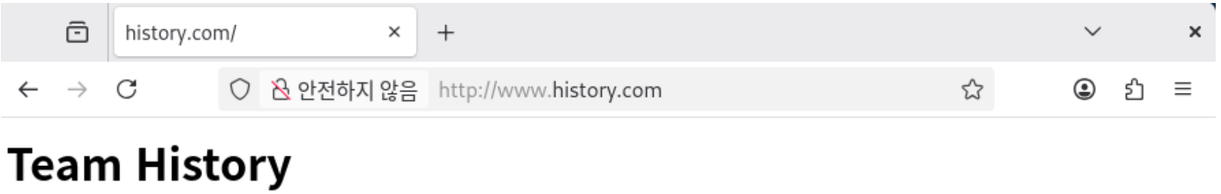
```
[root@project-b ~]# firewall-cmd --permanent --add-service=http
success
[root@project-b ~]# firewall-cmd --reload
success
[root@project-b ~]# firewall-cmd --list-services
cockpit dhcpv6-client dns http ssh
```

04. index.html 수정

```
[root@project-b ~]# cd /var/www/html
[root@project-b html]# vi index.html
<html>
        <body>
                <h1>Team History</h1>
        </body>
</html>
```

05. project에서 Web Server 접속

```
[root@project named]# curl www.history.com
<html>
        <body>
                <h1>Team History</h1>
        </body>
</html>
```



05 서버 구성 - FTP

01. FTP 설치 여부 확인 및 설치

```
[root@project-c ~]# rpm -qa vsftpd
[root@project-c ~]# dnf -y install vsftpd
```

02. 환경 설정

```
[root@project-c pub]# vi /etc/vsftpd/vsftpd.conf
12 anonymous_enable=YES
```

공개 FTP - 익명 로그인 활성화

03. 공유 디렉터리에 파일 생성

```
[root@project-c ~]# cd /var/ftp/pub
[root@project-c pub]# cp /boot/vmlinuz-5.* file1
```

파일 공유여부를 확인하기 위해 적당한
파일을 복사해서 가지고온다.

04. 서비스 시작

```
[root@project-c ~]# systemctl restart vsftpd
[root@project-c ~]# systemctl enable vsftpd
[root@project-c ~]# systemctl status vsftpd
● vsftpd.service - Vsftpd ftp daemon
     Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; prese>
     Active: active (running) since Wed 2026-01-14 15:13:25 KST; 17s ago
   Main PID: 31372 (vsftpd)
      Tasks: 1 (limit: 22762)
     Memory: 864.0K (peak: 1.1M)
        CPU: 3ms
     CGroup: /system.slice/vsftpd.service
             └─31372 /usr/sbin/vsftpd /etc/vsftpd/vsftpd.conf
```

05. 방화벽 허가

```
[root@project-c ~]# firewall-cmd --permanent --add-service=ftp
success
[root@project-c ~]# firewall-cmd --reload
success
[root@project-c ~]# firewall-cmd --list-services
cockpit dhcpv6-client ftp ssh
```

05 서버 구성 - FTP

01. project에서 ftp server에 접속

```
[root@project ~]# ftp ftp.history.com
Connected to ftp.history.com (192.168.111.200).
220 (vsFTPd 3.0.5)
Name (ftp.history.com:root): anonymous
331 Please specify the password.
Password:
230 Login successful.
```

익명 접속을 허용했기 때문에
Name에 anonymous를 입력하고
아무 패스워드를 입력한 뒤 접속한다.

02. 파일 확인 및 내려받기

```
ftp> cd pub
250 Directory successfully changed.
ftp> ls
227 Entering Passive Mode (192,168,111,200,146,77).
150 Here comes the directory listing.
-rwxr-xr-x    1 0        0        15092552 Jan 14 06:11 file1
226 Directory send OK.
ftp> get file1
local: file1 remote: file1
227 Entering Passive Mode (192,168,111,200,163,104).
150 Opening BINARY mode data connection for file1 (15092552 bytes).
226 Transfer complete.
15092552 bytes received in 0.126 secs (119693.82 Kbytes/sec)
```

03. 내려받은 파일 확인

```
[root@project ~]# ls -l
합 계  14744
-rw-------. 1 root root     1108  1월  7 15:30 anaconda-ks.cfg
-rw-r--r--. 1 root root 15092552  1월 14 15:23 file1
```

파일은 FTP를 접속했을 당시 디렉터리에 내려받는다.

05 서버 구성 - NFS Server

Rocky Linux

1. NFS 서버 설치 여부 확인 및 설치

```
[root@mail ~]# rpm -qa nfs-utils
nfs-utils-2.5.4-38.el9.x86_64
[root@mail ~]#
```

2. Exports 파일 수정

```
[root@mail ~]# vi /etc/exports
/share    *(rw,sync)
~
```

3. 공유 디렉토리 생성 및 권한 변경

```
[root@mail ~]# mkdir /share
[root@mail ~]# chmod 707 /share
```

4. 서비스 시작 및 활성화

```
[root@mail ~]# systemctl restart nfs-server
[root@mail ~]# systemctl enable nfs-server
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /usr/lib/sys
vice.
[root@mail ~]# systemctl status nfs-server
● nfs-server.service - NFS server and services
     Loaded: loaded (/usr/lib/systemd/system/nfs-server.service; enabled; preset: disabled)
    Drop-In: /run/systemd/generator/nfs-server.service.d
             └order-with-mounts.conf
     Active: active (exited) since Mon 2026-01-12 16:32:35 KST; 11s ago
```

5. CLI를 통한 방화벽 허가

```
[root@mail ~]# firewall-cmd --permanent --add-service=nfs
success
[root@mail ~]# firewall-cmd --reload
success
[root@mail ~]# firewall-cmd --list-services
cockpit dhcpv6-client dns imap nfs pop3 smtp ssh
```

6. 클라이언트측 공유 디렉토리 생성 및 마운트

```
[rocky@project-b ~]$ mkdir myShare
[rocky@project-b ~]$ ls
myShare  공개  다운로드  문서  바탕화면  비디오  사진  서식  음악
```

```
[rocky@project-b ~]$ su -c 'mount -t nfs 192.168.111.100:/share myShare'
암호:
```

```
[rocky@project-b ~]$ su -
암호:
[root@project-b ~]# vi /etc/fstab
/dev/mapper/rl-home     /home                   xfs     defaults        0 0
/dev/mapper/rl-swap     none                    swap    defaults        0 0

192.168.111.100:/share  /home/rocky/myShare     nfs     defaults        0 0
```

7. 클라이언트측 재부팅 후 공유파일 확인

```
[rocky@project-b ~]$ ls -l /home/rocky/myShare/
합계 14740
-rwxr-xr-x. 1 root root 15092552  1월 12 16:31 vmlinuz-5.14.0-611.5.1.el9_7.x86_64
[rocky@project-b ~]$ touch  /home/rocky/myShare/NFStest.txt
[rocky@project-b ~]$ ls -l /home/rocky/myShare/
합계 14740
-rw-r--r--. 1 rocky rocky        0  1월 12 16:52 NFStest.txt
-rwxr-xr-x. 1 root  root  15092552  1월 12 16:31 vmlinuz-5.14.0-611.5.1.el9_7.x86_64
```

05 서버 구성 - Samba

Rocky Linux

01. Samba 설치

```
[root@project ~]# dnf -y install samba samba-common samba-client
```

02. 공유 디렉터리 권한 부여

```
[root@project ~]# mkdir /sbhistory
[root@project ~]# groupadd sambaGroup
[root@project ~]# chgrp sambaGroup /sbhistory
[root@project ~]# chmod 770 /sbhistory
[root@project ~]# usermod -G sambaGroup rocky
[root@project ~]# smbpasswd -a rocky
New SMB password:
Retype new SMB password:
```

- 공유할 디렉터리를 생성한다.
- 공유 전용 그룹을 생성한다.
- 공유 디렉터리의 소유그룹을 변경한다.
- 디렉터리 권한을 제한한다. (소유자, 그룹)
- rocky를 그룹에 추가해 디렉터리 접근권한을 부여한다.
- rocky에게 smbpasswd를 부여한다.

03. 환경설정

```
[root@project ~]# vi /etc/samba/smb.conf
```

```
[global]
        workgroup = WORKGROUP
        security = user
        unix charset = UTF-8
        map to guest = Bad User
```

```
[sbhistory]
        path = /sbhistory
        writable = yes
        guest ok = no
        create mode = 0777
        directory mode = 0777
        valid users = @sambaGroup
```

04. 설정파일 오류 확인

```
[root@project ~]# testparm
Load smb config files from /etc/samba/smb.conf
Loaded services file OK.
Weak crypto is allowed by GnuTLS (e.g. NTLM as a compatibility fallback)

Server role: ROLE_STANDALONE

Press enter to see a dump of your service definitions
```

05 서버 구성 - Samba

Rocky Linux

05. 서비스 시작

```
[root@project ~]# systemctl restart smb nmb
[root@project ~]# systemctl enable smb nmb
```

06. 방화벽 허가

```
[root@project ~]# firewall-cmd --permanent --add-service=samba
success
[root@project ~]# firewall-cmd --permanent --add-service=samba-client
success
[root@project ~]# firewall-cmd --reload
success
[root@project ~]# firewall-cmd --list-services
cockpit dhcpv6-client dns imap mysql nfs pop3 samba samba-client smtp ssh
```

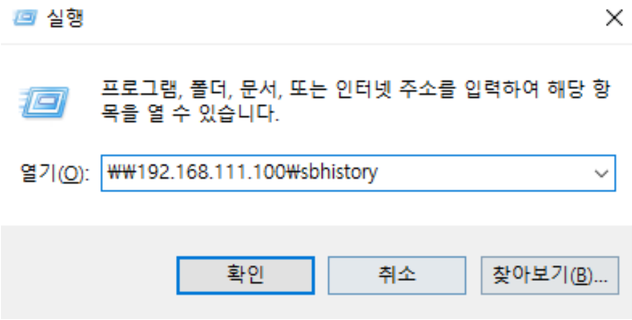
07. SELinux 설정

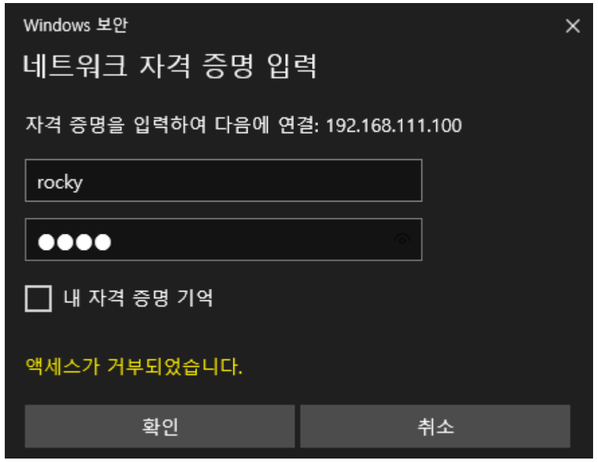
```
[root@project ~]# setsebool -P samba_enable_home_dirs on
[root@project ~]# chcon -R -t samba_share_t /sbhistory
```

08. 테스트용 파일 생성

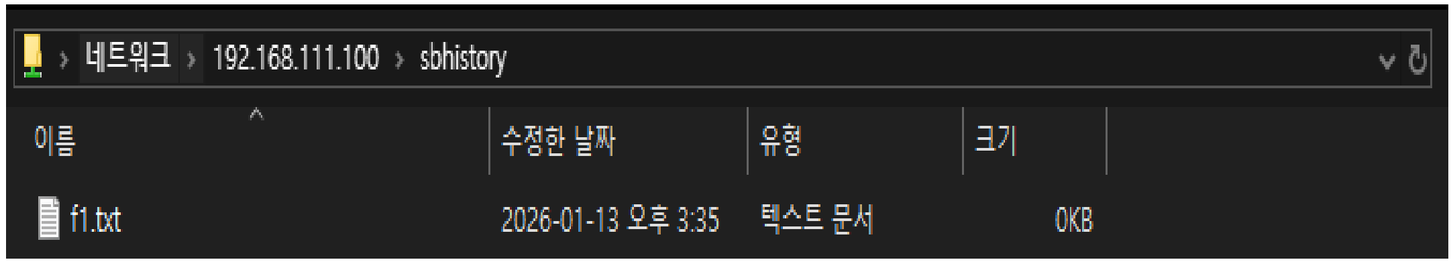
```
[root@project ~]# touch /sbhistory/f1.txt
[root@project ~]# ls -l /sbhistory
합 계  0
-rw-r--r--. 1 root root 0  1월  13 15:35 f1.txt
```

09. Windows Test





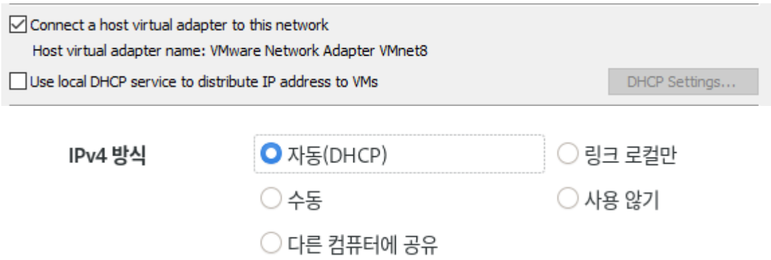
09. 공유된 파일 확인



05 서버 구성 - DHCP

Rocky Linux

01. 사전 작업



- VMware DHCP 체크 해제
- project-c의 네트워크를 DHCP로 지정

02. DHCP-Server 설치

```
[root@project ~]# dnf -y install dhcp-server
```

- project에 dhcp-server을 설치한다.

03. 환경설정

```
[root@project ~]# vi /etc/dhcp/dhcpd.conf
```

```
ddns-update-style interim;
subnet 192.168.111.0 netmask 255.255.255.0 {
        option routers 192.168.111.2;
        option subnet-mask 255.255.255.0;
        range dynamic-bootp 192.168.111.201 192.168.111.210;
        option domain-name-servers 8.8.8.8;
        default-lease-time 10000;
        max-lease-time 50000;
```

04. 서비스 시작

```
[root@project ~]# systemctl restart dhcpd
[root@project ~]# systemctl enable dhcpd
```

05. 방화벽 허가

```
[root@project ~]# firewall-cmd --permanent --add-service=dhcp
success
[root@project ~]# firewall-cmd --reload
success
[root@project ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns imap mysql nfs pop3 samba samba-client smtp ssh
```

05. 변경된 IP Address 확인

```
[root@project-c ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:f9:b9:52 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.111.201/24 brd 192.168.111.255 scope global dynamic noprefixroute ens160
       valid_lft 9972sec preferred_lft 9972sec
    inet6 fe80::20c:29ff:fef9:b952/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

- 환경설정에서 지정한 IP 범위의 첫번째 주소인 201번이 자동 배정되었다.

- DHCP는 브로드캐스트이기 때문에 같은 네트워크에 포함되어있으면 IP주소를 배정받을 수 있다.

05 서버 구성 - Mail Server

Rocky Linux

01. Mail서버 설치 여부 확인 및 설치

```
[root@project ~]# rpm -qa sendmail
[root@project ~]#
```

```
[root@history ~]# dnf -y install sendmail sendmail-cf dovecot
```

02. 호스트 이름 변경

```
[root@history ~]# hostnamectl set-hostname mail.history.com
```

```
[root@project ~]# vi /etc/hosts
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6

192.168.111.100      mail.hisory.com
~
```

```
[root@project ~]# vi /etc/mail/local-host-names
# local-host-names - include all aliases for your machine here.

mail.histroy.com
~
```

```
[root@project ~]# vi /etc/sysconfig/network
# Created by anaconda

HOSTNAME=mail.history.com
~
```

03. Named 디렉토리로 이동

```
[root@history ~]# cd /var/named
[root@history named]# pwd
/var/named
[root@history named]#
```

04. history.com.db 수정

```
[root@history named]# vi history.com.db
$TTL    3H
@       SOA     @       root.   ( 2     1D      1H      1W      1H )
        IN      NS      @
        IN      A       192.168.111.100
        IN      MX      10      mail.history.com

www     IN      A       192.168.111.150
ftp     IN      A       192.168.111.150
mail    IN      A       192.168.111.100
~
```

05. 오류검사

```
[root@history named]# named-checkconf
[root@history named]# named-checkzone history.com history.com.db
zone history.com/IN: history.com/MX 'history.com.history.com' has no address records (A or AAAA)
zone history.com/IN: loaded serial 2
OK
[root@history named]#
```

06. sendmail-cf 수정

```
[root@history ~]# vi /etc/mail/sendmail.cf
 85 Cwhistory.com
 86 # file containing names of hosts for which we receive email
 87 Fw/etc/mail/local-host-names
 268 O DaemonPortOptions=Port=smtp, Name=MTA
 269
```

05 서버 구성 - Mail Server

Rocky Linux

07. access 수정

```
[root@history ~]# vi /etc/mail/access
Connect:localhost.localdomain                   RELAY
Connect:localhost                               RELAY
Connect:127.0.0.1                               RELAY

192.168.111                                     RELAY
history.com                                     RELAY
~
```

08. Makemap hash

```
[root@history ~]# makemap hash /etc/mail/access < /etc/mail/access
```

09. dovecot 수정

```
[root@history ~]# vi /etc/dovecot/dovecot.conf
23 # Protocols we want to be serving.
24 protocols = imap pop3 lmtp submission
25
29 # edit conf.d/master.conf.
30 listen = *, ::
31
32 # Base directory where to store runtime data.
33 base_dir = /var/run/dovecot/
34
```

10. ssl 수정

```
[root@history ~]# vi /etc/dovecot/conf.d/10-ssl.conf
7 # plain imap and pop3 are still allowed for local connections
8 ssl = yes
9
```

11. mail 수정

```
[root@history ~]# vi /etc/dovecot/conf.d/10-mail.conf
24 #   mail_location = maildir:~/Maildir
25 mail_location = mbox:~/mail:INBOX=/var/mail/%u
26 #   mail_location = mbox:/var/mail/%d/%1n/%n:INDEX=/var/indexes/%d/%1n/%n
120 # mailboxes, or lm. s /secret/shared/box ~/mail/mybox would allow reading it).
121 mail_access_groups = mail
122
165 # methods. NFS users: flock doesn't work, remember to change mmap_disable.
166 lock_method = fcntl
167
```

12. 서비스 시작 및 활성화

```
root@history named]# systemctl restart named
root@history named]# systemctl enable named
root@history named]# systemctl status named
 named.service - Berkeley Internet Name Domain (DNS)
    Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: disabled)
    Active: active (running) since Mon 2026-01-12 15:17:47 KST; 16s ago
```

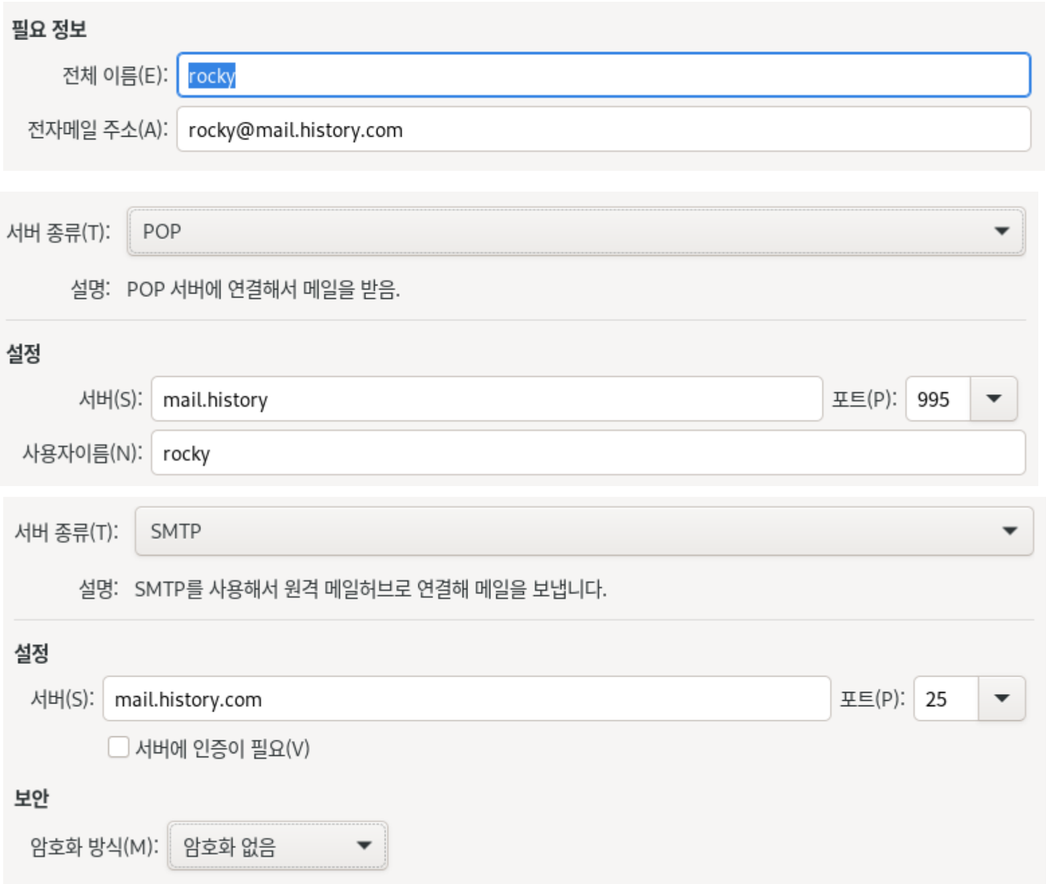
13. CLI를 통한 방화벽 개방

```
[root@history named]# firewall-cmd --permanent --add-service=smtp
success
[root@history named]# firewall-cmd --permanent --add-service=pop3
success
[root@history named]# firewall-cmd --permanent --add-service=imap
success
[root@history named]# firewall-cmd --reload
success
[root@history named]# firewall-cmd --list-services
cockpit dhcpv6-client dns imap pop3 smtp ssh
```

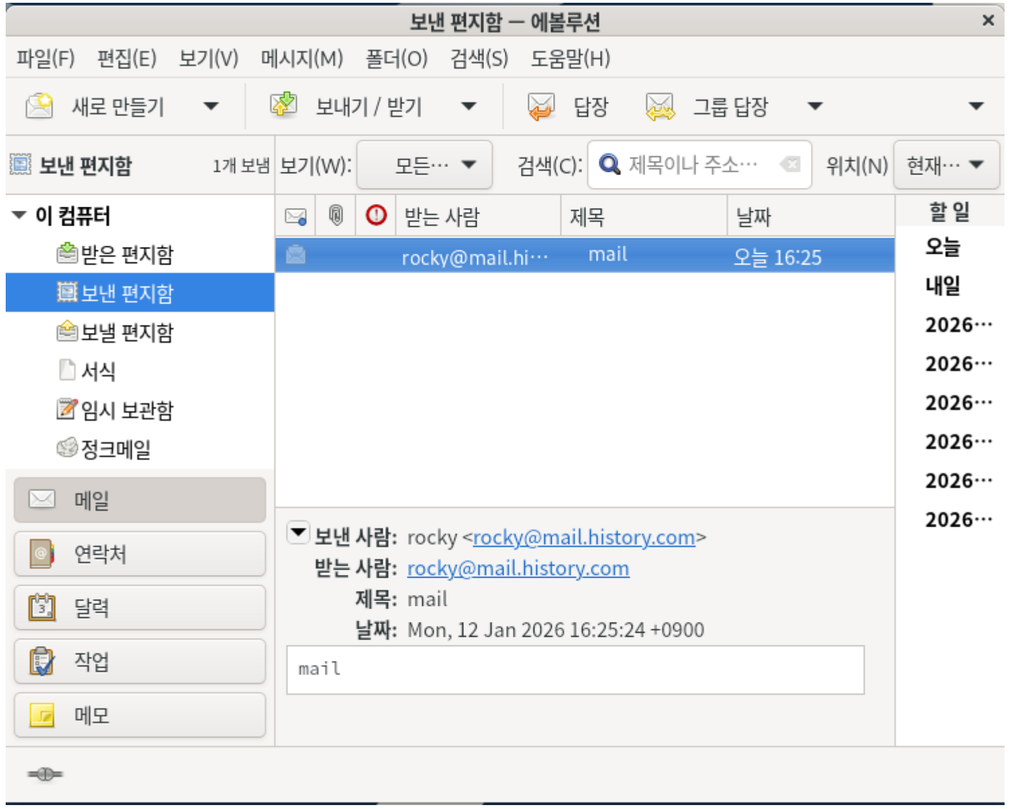
05 서버 구성 - Mail Server

Rocky Linux

14. 에볼루션 등록



15. 메일발송



05 서버 구성 - MariaDB Server

Rocky Linux

1. MariaDB 설치 여부 확인 및 설치

```
[root@mail ~]# rpm -qa mariadb-server
[root@mail ~]#
[root@mail ~]# dnf install mariadb-server
```

2. 서비스 시작 및 활성화

```
[root@mail ~]# systemctl restart mariadb
[root@mail ~]# systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.servi
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.serv
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/sy
[root@mail ~]# systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-01-12 10:57:55 KST; 13s ago
```

3. CLI를 통한 방화벽 개방

```
[root@mail ~]# firewall-cmd --permanent --add-service=mysql
success
[root@mail ~]# firewall-cmd --reload
success
[root@mail ~]# firewall-cmd --list-services
cockpit dhcpv6-client dns imap mysql nfs pop3 smtp ssh
[root@mail ~]#
```

4. DB 비밀번호 설정

```
[root@mail ~]# mysqladmin -u root password '1234'
```

5. DB 접속

```
[root@mail ~]# mysql -h localhost -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 4
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

6. DB 접속 권한 설정

```
MariaDB [(none)]> grant all on *.* to root@'%' identified by '1234';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.169 sec)

MariaDB [(none)]>
```

7. DB 생성

```
MariaDB [(none)]> CREATE DATABASE history CHARACTER SET utf8;
Query OK, 1 row affected (0.000 sec)
```

05 서버 구성 - MariaDB Server

Rocky Linux

8. DB 조회

```
MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| history            |
| information_schema |
| mysql              |
| performance_schema |
+--------------------+
4 rows in set (0.000 sec)
```

9. 클라이언트측 마리아DB 설치

```
[rocky@project-b ~]$ su -
암호:
[root@project-b ~]# dnf -y install mariadb
```

10. 클라이언트측 접속

```
[root@project-b ~]# mysql -h 192.168.111.100 -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 8
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

10. 클라이언트측 DB조회

```
MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| history            |
| information_schema |
| mysql              |
| performance_schema |
+--------------------+
4 rows in set (0.002 sec)
```

Rocky Linux

감사합니다

HISTORY